

Shaunak Bhandarkar

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Education

2024– **Ph.D. Candidate**, Neuroscience, Princeton University
 Advisors: Dr. Jonathan Pillow and Dr. Boris Hanin

2020-24 **B.S. with Honors**, Mathematics and Symbolic Systems, Stanford University
 Graduated with Distinction (GPA: 4.07)

Spring
2023 Visiting Student, Program in Mathematics, Oxford University

Interests: Neural Networks, Machine Learning Theory, Feature Learning, Learning Rules

Awards & Honors

2026-2029 National Defense Science and Engineering Graduate (NDSEG) Fellowship

2025-26 Princeton Quantitative Neuroscience Training Program Trainee ([NIH T32 Grant](#))

2024 Princeton Centennial Fellowship in the Natural Sciences and Engineering

2024 NSF Graduate Research Fellowship (NSF GRFP) Honorable Mention

2024 Phi Beta Kappa Honor Society

2024 Stanford George Pólya Honors Thesis Prize

2024 Stanford Mathematics Distinguished Service Award

Skills

Programming Languages: Python (PyTorch, JAX), C++, MATLAB, Java

Tools/Frameworks: Claude Code, Git, High-Performance Computing Clusters, LaTeX

Research Experience

2025– **Princeton Neuroscience Institute**, Princeton, NJ
 Graduate Researcher. Advisors: Dr. Jonathan Pillow and Dr. Boris Hanin

- Studying how gradient descent shapes task-relevant representations in neural networks and when these features produce successful learning; designing approaches to infer which learning rules best account for real animal learning.

2024–25 **Princeton Neuroscience Institute**, Princeton, NJ
 Graduate Rotation Student. Advisor: Dr. Jonathan D. Cohen

- Studied how memory/attention-based neural network architectures learn task-relevant representations, implicitly prioritizing certain task features over others.

2022–24 **Stanford University**, Stanford, CA
 Undergraduate Researcher. Advisor: Dr. James L. McClelland

- Developed biologically plausible neural network architectures for achieving the same functional outcomes as contemporary memory- and attention-based AI systems (e.g. Transformer-based language models), suggesting how the human mind may support similar complex computations as its AI counterparts.

2023–24

Stanford University, Stanford, CA

Undergraduate Honors Thesis. Advisor: Dr. Scott W. Linderman

- Investigated how neuromodulation adjusts low-rank recurrent neural network dynamics in solving sequence-based tasks, suggesting the role of neuromodulators in shaping flexible neural computation.

Publications and Abstracts

Peer-reviewed Conference Proceedings

1. Costacurta, J. C., **Bhandarkar, S.**, Zoltowski, D. M. & Linderman, S. *Structured flexibility in recurrent neural networks via neuromodulation* The Thirty-eighth Annual Conference on Neural Information Processing Systems 2024. [\[paper\]](#).

Conference Abstracts and Workshop Papers

1. Costacurta, J., **Bhandarkar, S.**, Zoltowski, D. & Linderman, S. *Neuromodulated Recurrent Neural Networks*. Computational and Systems Neuroscience (COSYNE). 2024. [\[poster\]](#).
2. **Bhandarkar, S.** & McClelland, J. *Sequential Learning and Retrieval in a Sparse Distributed Memory: The K-winner Modern Hopfield Network*. NeurIPS Associative Memory & Hopfield Networks. 2023. [\[manuscript + oral presentation\]](#).
3. **Bhandarkar, S.** & McClelland, J. *Sequential Learning and Retrieval in a Sparse Distributed Memory: The K-winner Modern Hopfield Network*. Conference on Cognitive Computational Neuroscience. 2023. [\[manuscript\]](#).

Manuscripts in Preparation or Under Review

1. **Bhandarkar, S.** & Hanin, B. *Investigating the Early Signatures of Feature Learning in Deep Linear Networks*. 2026. [in preparation; [poster](#)].
2. **Bhandarkar, S.** & Pillow, J. *Inferring learning rules in deep neural network architectures from animal learning data*. 2026. [Under review for NeurIPS 2026].

Preprints & Miscellaneous Works

1. **Bhandarkar, S.** & McClelland, J. *Neural Computation Without Slots: Steps Towards Biologically Plausible Memory and Attention in Natural and Artificial Intelligence*. ArXiv. 2025. [\[paper\]](#).
2. **Bhandarkar, S.** *Neuromodulated Low-Rank Recurrent Neural Networks: A Mechanistic Account of Neuromodulation in Shaping Flexible Neural Dynamics*. 2024. [\[undergraduate honors thesis\]](#).
3. **Bhandarkar, S.** *On the Density Theorem of Cebotarev*. Ball State Undergraduate Mathematics Exchange. 2021. [\[paper\]](#).

Presentations

Talks

1. **Bhandarkar, S.** *"Sequential Learning and Retrieval in a Sparse Distributed Memory: The K-winner Modern Hopfield Network."* Bay Area Memory Meeting (BAMM!) (Stanford, CA). 2023.

Posters

1. **Bhandarkar, S.** & Cohen, J. *Attention-based Memory Supports Learning of Semantically Rich Representations.* Princeton Machine Learning Theory Summer School. 2025. [\[poster\]](#).
2. **Bhandarkar, S.,** Chou, C.-N. & Chung, S. *The Geometry of Learning: How One Gradient Step Alters Latent Manifold Structure* Flatiron Institute Summer Research Poster Session (New York, NY). 2023. [\[poster\]](#).

Teaching

Princeton University

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| 2026 | Teaching Assistant, Introduction to Cognitive Neuroscience (NEU 202) <ul style="list-style-type: none">• Introducing undergraduates to topics in cognitive neuroscience; commitments include holding precepts for discussing papers and grading. |
| 2025 | Teaching Assistant, Mathematical Tools for Neuroscience (NEU 314) <ul style="list-style-type: none">• Introducing undergraduate neuroscience majors to quantitative neuroscience topics (programming, linear algebra, dynamical systems, probability); commitments include holding review/help sessions, creating assignments and exams, and grading. |

Community Involvement

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| 2023-24 | Undergraduate Math Tutor for Stanford Math Department , Stanford, CA <ul style="list-style-type: none">• Helped undergraduates learn topics and solve problems in multivariate calculus, linear algebra, and differential equations, as part of the Stanford Undergraduate Math Organization (SUMO). |
| 2022-24 | Student Peer Advisor for Stanford Math Department , Stanford, CA <ul style="list-style-type: none">• Helped undergraduates navigate courses, research opportunities, degree plans, and their math-related interests. Additionally organized guest talks by faculty and graduate students in the math department. |
| 2016-24 | Head Instructor at Special Needs Aquatic Program (SNAP) , Palo Alto, CA <ul style="list-style-type: none">• Helped kids with special needs exercise their muscles and have fun in the Palo Alto YMCA pool. Led group swim activities and recruited new Stanford volunteers into the program. |
| Summer 2022 | Teaching Assistant for Stanford University Math Camp (SUMaC) , Stanford, CA <ul style="list-style-type: none">• Helped advanced 11th and 12th grade students learn topics and solve problems in abstract algebra and algebraic topology. |